

Stanford Undergraduate Research Opportunity Discovery

User Discovery Documentation

Project: Stanford Research Opportunity Matching Platform

Principal Investigator: Lanon (Freshman, Electrical Engineering, Robust Systems Group)

Date: March 2026

Purpose: Project Validation

Total Interviews: 5 students + 1 self-reflection

Validation Status:  VALIDATED

Executive Summary

Problem Statement

Research access at Stanford operates through hidden pipelines (specific intro seminars → summer programs → honors tracks, or department-specific relationships) or inefficient mass outreach (cold-emailing dozens of professors with low response rates). Students who don't "stumble in" through these pipelines face real friction:

- No central, searchable database of labs actively accepting underclassmen
- Departmental listings (CURIS, EE REU pages, Biology faculty pages) are siloed, uneven, and sometimes outdated
- International students face invisible citizenship/visa eligibility barriers
- Students who switch departments lose their lab connections and start over
- Freshmen and sophomores are either unaware that research is accessible to them or don't know how to begin

Key Finding

4 out of 6 people (67%) confirmed the problem and would use a centralized solution, including the founder. Even students who successfully found research described the process as luck-based, relationship-gated, or only possible because they happened into the right class or knew the right person.

Validation Strength: 9/10

-  Founder experiences problem personally (strong founder-market fit)
-  Multiple user segments validated (freshmen, sophomores, juniors, master's students)

- ☒ Consistent pain points across departments (CS, EE, Cell Bio)
- ☒ Specific, actionable feature requests from real users
- ☒ Problem confirmed beyond a single department
- ☒ Even successful students validate the inefficiency of the current system

Recommended Action

Build a manual MVP immediately (this week), test with founder + 10–15 students, iterate based on real usage, then build the technical platform.

Research Methodology

Interview Framework

Semi-structured interviews with open-ended questions designed to surface actual behavior and pain points rather than confirm predetermined assumptions.

Core Questions:

1. "Have you tried to find research opportunities at Stanford?"
 - If yes → "How did you do it? What was hard about it?"
 - If no → "Why not? Is it something you want?"
2. "Do you know which labs or professors actively take freshmen or sophomores?"
 - If no → "How would you find out?"
3. "Have you looked at CURIS, the EE REU pages, or your department's research site?"
 - If yes → "Was it helpful? Did you find what you needed?"
4. "If there was a searchable database of Stanford labs currently taking underclassmen, would you use it?"
 - Listen for genuine interest vs. polite agreement
5. "What would make you ACTUALLY use it vs. just bookmark and forget?"

Interview Duration: 20–30 minutes per participant, conducted informally in-person or via video call

Sample Selection: Purposive sampling targeting multiple class years (freshman, sophomore, junior, master's), multiple majors (CS, EE, Cell Biology), and both research-active and non-research-active students.

Limitations:

- Small sample size (n=6)
 - Underrepresentation of humanities and social science students
 - No formal interviews with faculty yet
 - Self-selection bias (students willing to talk about research)
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Detailed Interview Results

Person 1: CS Major, Freshman

Profile: Builder-focused, strong independent project track record, has not pursued academic research

Key Findings:

- Not interested in academic research — prefers building products and competing in hackathons
- Acknowledged that Stanford's research landscape feels opaque even from a distance
- Didn't know about CURIS or the year-round CS project board until this interview
- Would not pursue unpaid or purely theoretical research; unclear even if paid
- Heavy course and extracurricular load limits bandwidth

Direct Quotes:

- "I feel like research is for grad students honestly. I don't even know where I'd start."
- "I've heard CURIS but I thought that was only for upperclassmen. Is it not?"
- "I'd probably just rather build my own stuff than help someone else's project."

User Classification: ❌ Not a target user

Value to Research:

- Validated that CURIS's freshman eligibility is not well-known, even among motivated CS students
 - Identified that the builder vs. researcher divide is real — not all STEM students want academic lab work
 - Suggested paid/project-based framing as a potential hook even for builder-types
 - Represents a significant segment (~25–30%) of Stanford CS students
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Person 2: CS Major, Freshman — Currently Doing Materials Science Research

Profile: Found a research position in a Stanford materials science lab early in freshman year; got in through a personal connection made at a department event

Key Findings:

- Stumbled into research by attending an MSE department mixer where a PhD student mentioned an opening
- Describes the process as "completely random" — he was not actively looking for MSE research
- Not aware of any formal mechanism that would have led him there otherwise
- His lab is doing work on novel semiconductor materials for energy-efficient computing
- Lab is friendly to underclassmen but he doesn't know if they're formally "open" — they never advertised the spot
- Would not have found this through a website; it happened through a hallway conversation

Direct Quotes:

- "I only found out because I went to this mixer kind of on a whim and a PhD student mentioned they were looking for undergrads."
- "The lab doesn't really post anywhere. I think they just ask people they meet."
- "I didn't even know I wanted to do materials science until I was literally already in the lab."
- "If there was a database I probably would have found something better matched to what I actually care about."

User Classification: ⚠️ Partially Validated — has research but confirms the discovery system is broken

Value to Research:

- Critical validation that successful students still describe discovery as "completely random" or luck-based
- Revealed that many labs recruit exclusively through personal networks, not formal channels — exactly the gap a platform would fill
- MSE department has a formal 10-week summer research program open to all years including freshmen, but he wasn't aware of it
- His quote about "better matched" is key: even when students find research by accident, it may not align with their interests

Insights:

- Lab openings exist but aren't advertised — information asymmetry, not actual scarcity
- Department events (mixers, seminars, intro seminars) are informal hiring pipelines
- Students outside these social touchpoints are completely shut out

Person 3: Master's Student in EE, Robust Systems Group (RSG)

Profile: Graduate student (MS) in Prof. Mitra's Robust Systems Group, which works on robust computing, NanoSystems, EDA, and brain-computer interfaces. Has significant insight into how the group works and what undergrads experience there.

Key Findings:

- RSG actively lists itself as open to undergrads — the website explicitly says "We are always looking for talented new undergraduate students" with an email to send a CV to
- In practice, undergrad onboarding varies — some arrive through formal email outreach, others through taking EE 108 (Digital Systems Design, taught by Prof. Mitra) and catching the professor's attention
- The group's work spans hardware, software, and neuroscience, making it accessible from multiple angles (EE, CS, even bio)
- Masters students in the group often serve as de facto mentors for undergrad researchers
- Freshmen CAN join, but benefit from having some EE or CS fundamentals — EE 108 is often the entry point
- Undergrad projects are real and non-trivial — the group works on carbon nanotube computers, 3D NanoSystems, and silicon-level reliability testing

Direct Quotes:

- "We do take undergrads. The website says so and I've seen it happen. But I don't think most undergrads know how to find us or what we even do."
- "If someone took EE 108 and showed genuine interest, Prof. Mitra would remember them. That's probably the most reliable path."
- "A lot of the undergrads who show up don't realize how broad the research actually is — it's not just circuits. There's neuro, there's nano, there's systems."
- "The email on the website works. But you need to have something real to say — just saying you're interested isn't enough."

User Classification:  Insider validator — provides supply-side (lab) perspective

Value to Research:

- Confirmed that a well-known, prestigious lab is open to undergrads but is not effectively reaching them
- Identified EE 108 as a hidden pipeline into RSG — a "right class" dynamic similar to Howard's General Bio Lab finding
- Highlighted that a cold email to RSG works IF the student demonstrates concrete skills or interest — which requires knowing the lab exists and what they do
- Underscored the gap: RSG's openness is technically public, but practically invisible to most freshmen

Person 4: Sophomore, Cell Biology — Doing Research at Stanford Medicine

Profile: Cell biology sophomore who found a Stanford Medicine lab through a combination of Bio-X program materials and a direct email to a professor whose lab website mentioned undergrad openings

Key Findings:

- Did find research, but the process took most of freshman year and required significant self-directed research
- Used Community Academic Profiles (a Stanford Medicine tool to search faculty by keyword) — describes it as "useful but overwhelming"
- Sent 14 emails; received 3 responses; had 2 conversations; accepted 1 offer
- Lab is working on cellular signaling pathways relevant to cancer biology — wet lab, paid position through Bio-X USRP summer program, continued during academic year
- Didn't know about Bio-X USRP until she found a flyer in the bio building hallway — not through any official orientation
- Described biggest pain points as: not knowing who was actively taking students, not knowing what experience was required, and not knowing if spots were paid

Direct Quotes:

- "I sent like 14 emails and most professors just didn't reply. It felt really demoralizing."
- "The Community Academic Profiles tool is fine but there's no way to tell who's actually accepting students right now versus whose lab is full."
- "I found out about Bio-X from a flyer. A flyer. Why is that not in orientation?"
- "If there was a filter for 'wet lab vs. dry lab' and 'currently accepting' I would have saved so much time."
- "Some of the lab websites hadn't been updated in like two years. I emailed a professor about a project that was already published and done."

User Classification:  PRIMARY TARGET USER

Value to Research:

- First hard validation of spray-and-pray email behavior and its demoralizing effect (14 emails, 3 replies)
- Confirmed outdated lab websites as a major pain point — directly parallel to Howard's finding
- Identified Community Academic Profiles as the closest existing tool but critically lacking real-time availability data
- Confirmed Bio-X USRP as an excellent pathway that is not effectively communicated to incoming students
- Generated the clearest feature requests: wet/dry lab filter, currently accepting (Y/N), paid/unpaid

Insights:

- Stanford has the infrastructure but not the UX — tools exist but are siloed, incomplete, or hard to discover

- Students outside Bio-X's formal network still struggle even when they're motivated and capable
 - Sending 14 emails is the current workaround — a platform could compress that to 2–3 targeted outreaches
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Person 5: Junior, EE — Doing Hardware Verification Research at Thambes Lab

Profile: Junior doing hardware verification and formal methods research. Got in through a recommendation from a PhD student he knew socially. Describes the work as technically interesting in principle but somewhat tedious in day-to-day execution.

Key Findings:

- Found the lab through a PhD student in his friend group — not through any official channel
- Lab works on hardware verification and formal methods for digital systems — important but niche
- Describes the day-to-day as "a lot of reading papers and running tools, not much building"
- Would not have found this lab independently — has no idea how a student without the social connection would do it
- Aware that many of his peers who want research are struggling to find any position at all
- Sees value in a platform but from a "quality matching" angle — he wants students to find labs they'll actually enjoy, not just any lab

Direct Quotes:

- "Honestly I kind of got lucky. My friend knew someone in the lab and that's how I found out they were looking."
- "The work is fine, but I think if I'd had more information upfront about what day-to-day in this lab looks like I might have chosen differently."
- "A lot of my friends are just emailing professors randomly and not hearing back. It seems really inefficient."
- "What would actually help is if there was like a description of what the work actually involves, not just the professor's research summary."
- "There's no way to know what a lab is like until you're in it."

User Classification:  SECONDARY TARGET USER — has research but validates system inefficiency and adds quality-matching insight

Value to Research:

- Critical new insight: discovery is only part of the problem — fit matters too. Students want to know what day-to-day research looks like before committing

- Reinforced the luck-based access pattern: social connections determine who gets research, not interest or ability
- Highlighted that formal methods/verification is a less visible but real area where undergrads can contribute — and where students have trouble finding labs
- Generated a new feature idea: "day-in-the-life" lab description beyond just the professor's research abstract

Insights:

- Students who find research through social networks may end up in labs that don't match their working style
 - Lab culture and day-to-day work type (building vs. reading/running tools vs. wet lab) matters and is not captured anywhere
 - Even satisfied researchers see systemic problems — platform is valuable beyond just struggling students
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Founder Self-Reflection: EE Freshman, Robust Systems Group

Profile: Freshman in EE, member of Prof. Mitra's Robust Systems Group (unusual for a freshman), also involved in Stanford Student Robotics as an electrical systems specialist on the Battle Bots team. Strong hands-on background: FPGA programming, graph theory projects, power distribution system design. Has interest in both deep technical research and applied engineering.

Key Findings:

- Got into RSG through proactive outreach combined with a strong technical background — not a standard path
- Describes it as "not exactly easy to figure out that this was even possible as a freshman"
- Did not know about CURIS, the year-round CS project board, or the EE REU pages before getting placed
- Knows that peers with equal or greater technical ability are struggling to find any research position
- Particularly attuned to the international student dimension — eligibility requirements for certain programs (e.g., some NSF-funded REUs) are not transparent
- Has a clear view of what a platform would need to contain given his insider position in an active lab
- Would use a searchable database personally for future opportunities and to help peers navigate

Direct Quotes:

- "I got in but I honestly think it was more about knowing to email in the right way than anything else. Most people don't know how to do that."
- "There's no freshman orientation for research. No one tells you that you can just email a professor and sometimes they say yes."

- "I know people in my dorm who are interested in research and have no idea where to start. They don't know CURIS exists, they don't know about Bio-X, they don't know any of it."
- "Some programs are only for US citizens and that's never written clearly anywhere. I've had to find that out the hard way."
- "If there was a page that just said: here are the labs, here's what they do, here's if they take freshmen, here's if international students are eligible — that would have saved me a lot."

User Classification:  PRIMARY TARGET USER + USER #1

Value to Research:

- Strongest validation: founder is currently navigating the exact problem being solved, from the inside of a lab
- Unique insight into international student eligibility gaps — a dimension missing from most existing resources
- Firsthand knowledge of what makes a successful cold email vs. a generic one — points to a platform feature (email templates or guidance)
- Confirmed that peer-to-peer knowledge sharing is currently how most students find research — platform would democratize that

Validation Summary Table

Person	Profile	Problem Exists?	Would Use Tool?	Key Insight
Person 1	CS Fresh, no research	 Not interested in academic research	No	"Research is for grad students" — freshmen don't know it's accessible; CURIS eligibility not widely known
Person 2	CS Fresh, materials science	 Confirmed but found research by luck	Would've used it	"Completely random" discovery; lab doesn't advertise; would have found better fit with a platform
Person 3	EE Master's, RSG	 YES — labs don't reach undergrads effectively	N/A (supply side)	RSG takes undergrads but most freshmen don't know; EE 108 is hidden pipeline
Person 4	Cell Bio Soph, Stanford Medicine	 YES — 14 emails, 3 replies, outdated websites	 YES	"Why is Bio-X not in orientation?"; needs wet/dry filter, currently accepting, paid/unpaid

Person 5	EE Junior, Thambes Lab	✓ YES — social network gated, tedious for peers	✓ YES	"Got lucky" via friend connection; wants day-in-the-life descriptions, not just research abstracts
Founder	EE Fresh, RSG, international	✓ YES — international barriers, no freshman roadmap	✓ YES	No orientation for research; citizenship requirements opaque; could help peers immediately

Validation Rate: 4/6 people (67%) confirmed problem and would use solution

Target User Profile: Research-motivated freshmen/sophomores without existing social connections to labs, particularly international students and those outside established pipelines (honors thesis tracks, department programs)

Problem Analysis

Core Problem (Validated)

Research access at Stanford is network-gated and fragmented. Success requires either:

- Stumbling into the right class, event, or social connection at the right time
- Performing an exhausting spray-and-pray email campaign with low returns
- Already having insider knowledge of which labs, programs, and portals exist

Root Causes

1. Fragmented Information Infrastructure Stanford has multiple research portals — CURIS (CS), EE REU pages, Bio-X USRP, Community Academic Profiles (Medicine), Biology faculty listings, SURP — but no central interface. Students must know these exist and navigate each separately. None of them surfaces real-time availability, lab type, or year eligibility clearly.

2. Hidden Institutional Pathways Certain classes reliably lead to research (EE 108 → RSG; General Biology Lab → Bio-X pathway; certain Intro Seminars → professor relationships) but this is never made explicit. Students outside these classes are structurally disadvantaged.

3. Inefficient Discovery Process Cold emails are the default fallback and they mostly don't work. Person 4 sent 14 emails for 3 responses. The Founder's email to RSG worked, but only because it was unusually well-crafted. Most students don't know how to write an effective outreach email, and without a description of what labs are looking for, they can't tailor their message.

4. Invisible Barriers for Specific Populations

- International students face citizenship requirements (NSF REU funding bars non-US citizens; some departmental programs require US permanent residency) that are not prominently disclosed. Students discover ineligibility after investing time.
- Students without pre-existing faculty or grad student relationships are at a systematic disadvantage vs. those with social capital.
- Students who change departments or arrive without a declared major lose departmental pipeline access.

5. Quality Mismatch Problem (Unique to Stanford — Not Raised at Howard) Even students who find research may end up in labs that don't fit their working style (theoretical vs. hands-on, wet vs. dry lab, verification vs. systems building). Because discovery is hard enough, students take the first offer. A platform that includes day-to-day work descriptions could improve fit, not just access.

Stanford Labs Actively Welcoming Underclassmen

The following labs and programs have confirmed openness to freshmen and/or sophomore researchers (based on public documentation and interview-sourced intelligence):

1. Robust Systems Group (RSG) — Prof. Subhasish Mitra

Department: Electrical Engineering & Computer Science

Website: rsg.stanford.edu

Research Focus: Robust computing, NanoSystems (carbon nanotube computers, 3D NanoSystems), Electronic Design Automation (EDA), brain-computer interfaces, and circuit reliability. Prof. Mitra's X-Compact technique is embedded in virtually every modern semiconductor manufacturing test process. Recent work includes the first carbon nanotube computer (NATURE cover) and the first resistive RAM-based computing systems.

Openness to Underclassmen:  Explicitly listed on website — "We are always looking for talented new undergraduate students." Email subh@stanford.edu with a short introduction and CV.

Best Entry Path: Take EE 108 (Digital Systems Design) and perform well, or cold email with a well-targeted, technically grounded message. Demonstrated FPGA, hardware design, or programming skills strongly valued.

International Students: Not listed as restricted — confirm directly.

Paid/Unpaid: Varies by project and funding; ask during initial outreach.

Lab Type: Hardware + computational (mixed); no wet lab.

2. Materials Science & Engineering Research Program

Department: MSE

Website: mse.stanford.edu/academics-admissions/undergraduate-program

Research Focus: Materials for energy-efficient computing, biomedical applications, nanotechnology, electronic and photonic materials, mechanical behavior of materials. Labs span experimental (fabrication, characterization, X-ray diffraction, nanomaterials) and computational approaches.

Openness to Underclassmen:  Formal 10-week summer program explicitly open to freshmen through seniors. "Students of all years ranging from freshman to seniors are given opportunities to work closely with a professor." Many continue during the academic year afterward.

Best Entry Path: Apply to the MSE Undergraduate Research Program in spring for summer. Alternatively, attend MSE department events — informal recruiting happens there (confirmed by Person 2).

International Students: Confirm per individual lab; summer program eligibility may vary.

Paid/Unpaid: Paid summer stipend through the formal program; academic year varies by lab.

Lab Type: Predominantly wet lab / fabrication; some computational.

3. Bio-X Undergraduate Summer Research Program (USRP)

Department: Interdisciplinary (Biology, Bioengineering, Chemistry, CS, Medicine, and more)

Website: biox.stanford.edu/research/undergraduate-research

Research Focus: Broadly interdisciplinary at the intersection of engineering, biology, and medicine. Labs span cancer biology, neuroscience, immunology, structural biology, synthetic biology, and computational biology. In Summer 2025, Bio-X USRP hosted 69 Stanford students across labs campus-wide.

Openness to Underclassmen:  Open to all undergraduate years. Participating students are matched with faculty mentors across departments. Students can continue in their lab during the academic year after the summer program. CS majors and EE majors have participated — not limited to biology students.

Best Entry Path: Apply to Bio-X USRP in late winter/early spring for the following summer. Alternatively, identify individual Bio-X affiliated faculty and email directly using the Community Academic Profiles tool (profiles.stanford.edu — searchable by keyword).

International Students: Confirm per individual lab; some positions funded through grants with citizenship requirements.

Paid/Unpaid: Paid (summer stipend); academic year component varies.

Lab Type: Predominantly wet lab; some computational and dry lab options.

4. CURIS Year-Round CS Projects

Department: Computer Science

Website: curis.stanford.edu

Research Focus: All areas of CS research — AI/ML, systems, HCI, theory, robotics, programming languages, and more. Faculty post projects throughout the year seeking undergrad research assistants for research units, hourly pay, or RA positions.

Openness to Underclassmen:  Explicitly open to freshmen and sophomores. PURE

(Paid Undergraduate Research Experience) subprogram specifically targets students with little or no prior research experience, designed as an on-ramp.

Best Entry Path: Browse curis.stanford.edu for year-round project listings. For PURE specifically, must have Federal Work Study eligibility (US citizens and permanent residents only — international students are ineligible for PURE funding but can still apply for lab positions through other channels).

International Students: ⚠️ PURE funding requires Federal Work Study — US citizens and qualifying permanent residents only. Non-PURE lab positions may still be available; confirm with individual faculty.

Paid/Unpaid: PURE positions are paid; other listings vary.

Lab Type: Computational / software; no wet lab.

5. Stanford Physics Undergraduate Research Program (SURP)

Department: Physics

Website: physics.stanford.edu/undergraduate/undergraduate-research

Research Focus: Experimental and theoretical physics across Stanford's Physics Department affiliates, which include SLAC, KIPAC (astrophysics/cosmology), and various condensed matter, particle physics, and quantum information groups.

Openness to Underclassmen: ✅ "We accept applications from current freshmen, sophomores, and juniors. For beginning students, research experience is not required." Priority for funding: freshmen and rising sophomores with no prior funding are in the second-highest priority tier.

Best Entry Path: Apply through the department's formal program. Funding supported by the Physics Department for at least one quarter per Physics/Engineering Physics major. Non-physics majors can still participate with a willing faculty advisor.

International Students: Program is open; confirm funding eligibility per quarter.

Paid/Unpaid: Funded positions available through the department program.

Lab Type: Experimental and computational; some overlap with EE for quantum/photonics work.

6. Parker Lab — Stanford Medicine (Cell Biology / Immunology)

Department: Stanford School of Medicine — Surgery / Transplantation

Website: med.stanford.edu/parkerlab/join-us/undergraduate-students.html

Research Focus: Immunology and transplantation biology. Undergrads in the Parker Lab typically pursue Human Biology or Biology majors and have gone on to publish papers, secure independent grants, and complete honors theses.

Openness to Underclassmen: ✅ Explicitly accepts undergraduate researchers year-round. Multiple participation pathways: academic year research (course credit, work-study, honors), paid summer fellowships (HB-REX, Bio-X, Amgen).

Best Entry Path: Email Dr. Parker and/or Desiree Dupes directly with a resume/CV and a statement of research interests. The lab website specifies exactly what to include. Minimum one-year commitment expected.

International Students: Confirm individual positions — some funding streams may have restrictions.

Paid/Unpaid: Both; paid options available through summer fellowship programs.

Lab Type: Wet lab (immunology, transplantation, in vivo work).

7. Stanford Cancer Institute / ACS Diversity in Cancer Research Program

Department: Stanford Cancer Institute (interdisciplinary)

Website: eso.stanford.edu/programs/undergraduate-students

Research Focus: Cancer research across biology, chemistry, bioengineering, and medicine. Program places students in Stanford Cancer Institute labs with supplementary scientific seminars and career development workshops.

Openness to Underclassmen:  Specifically targets underrepresented minority undergraduates at both Stanford and Bay Area community colleges/four-year institutions.

Best Entry Path: Apply through the Education & STEM Outreach office during the application window (typically winter quarter).

International Students: Confirm per application — program is focused on diversity and underrepresented students.

Paid/Unpaid: Funded by the American Cancer Society — includes stipend.

Lab Type: Wet lab and computational; depends on placement.

Feature Requirements (User-Driven)

Must-Have Features (MVP)

Feature	Requesting Users	Pain Point Addressed
Search by research area/keyword	Person 3, 4, 5, Founder	Students don't know which labs match their interests
Currently accepting students (Y/N/Seasonal)	Person 4, 5, Founder	Students waste time emailing full or closed labs
International student eligibility	Founder (unique insight)	~10–15% of Stanford undergrads face hidden eligibility barriers
Lab type (Wet/Dry/Computational/Hardware)	Person 4, 5	Working style fit matters; Person 5 might have chosen differently
Paid vs. unpaid	Person 4, Founder	Payment structure affects student decision-making

Experience required (none/some/significant)	Person 4, Founder	Students fear rejection for being "too junior"
Day-to-day work description	Person 5 (unique insight)	"Research summary" ≠ what the work actually involves
Professor/lab contact info	All	Streamlines outreach
Class year accepted (freshmen welcome Y/N)	Person 1, Founder	Freshmen don't know research is accessible to them
Hidden pipeline flag ("Best entry via EE 108")	Person 3, Founder	Surface institutional knowledge that currently lives only in social networks

Nice-to-Have Features (V2)

Feature	Purpose
Email templates by lab type	Reduce outreach friction; most students don't know how to write a good cold email
Student reviews / lab culture notes	"Prof. X is very hands-on with undergrads" vs. "you mostly run code independently"
Application deadlines	Students miss CURIS, Bio-X windows because they don't know they exist
Time commitment (hrs/week)	Students need to balance research with coursework
Recent publications / active projects	Know what the lab is working on RIGHT NOW, not 3 years ago
Cross-departmental tags	EE/CS overlap, Bio/ChEM-H overlap, Physics/EE quantum overlap
Summer vs. academic year availability	Bio-X is summer; CURIS has year-round; students need to know the difference

Competitive Analysis

What Currently Exists at Stanford

Resource	What It Does	Critical Gaps
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CURIS (curis.stanford.edu)	Lists CS year-round and summer research projects; organizes CURIS summer program	CS-only; PURE requires FWS eligibility (excludes international students); no lab-type or year filter
Bio-X USRP	10-week summer program placing undergrads in interdisciplinary bio labs	Application-based; no real-time browsing of who's currently accepting; discovery requires knowing Bio-X exists
Community Academic Profiles (Medicine)	Search Stanford Medicine faculty by name, department, or keyword	No availability filter; no undergrad-specific view; overwhelming for freshmen; often outdated
EE Department REU Pages	Lists some EE-specific research opportunities	Sparse; not comprehensive; not updated regularly
Biology Department Faculty Page	Lists faculty with research summaries	No accepting/not accepting filter; no year filter; no paid/unpaid info; no lab type
MSE Undergraduate Research Program	Formal 10-week summer program open to all years	Summer only; no academic-year companion; not well-known (Person 2 didn't know it existed)
Stanford SURP (Physics)	Formal program for physics majors	Physics/Eng. Physics only; not cross-departmental
Undergraduate Research & Independent Projects Office	General guidance and grants	No lab directory; high-level advice only

What Doesn't Exist at Stanford:

- ❌ Cross-departmental, searchable database of labs currently accepting underclassmen
- ❌ International student eligibility filtering
- ❌ Real-time availability (accepting now vs. waitlist vs. closed)
- ❌ Day-to-day work descriptions (builds/hardware vs. reads papers vs. runs experiments)
- ❌ Hidden pipeline surfacing ("takes students from EE 108"; "go to MSE mixers")
- ❌ Any single entry point that works across CS, EE, Biology, Medicine, and Physics

Validation Summary

Overall Assessment: VALIDATED — PROCEED TO BUILD

Validation Strength: 9/10

Evidence:

- ☒ 4/6 people (67%) confirmed problem and would use solution
- ☒ Founder experiences problem personally (strongest possible validation)
- ☒ Even successful students (Person 2, Person 5) describe discovery as luck-based
- ☒ Specific, actionable feature requests from real users
- ☒ Problem confirmed across CS, EE, Cell Biology
- ☒ Unique insights from international student perspective
- ☒ Supply-side validation from Master's student (labs are open but students don't know)
- ☒ Quality-matching gap identified (Person 5) — a dimension the Howard analysis didn't surface

Remaining Unknowns to Address This Week:

1. Will students actually email professors after finding them? (Test: manual MVP, track behavior)
 2. Will professors/labs agree to be listed? (Test: email 2–3 lab contacts this week)
 3. Does the problem extend to Humanities and Social Science researchers? (Optional: interview 1–2 non-STEM students)
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Recommended Next Steps

Phase 1: Final Pre-Build Validation (This Week — 3–4 hours)

Talk to 1–2 faculty or lab contacts:

- Purpose: Confirm labs will participate; understand professor perspective on undergrad recruitment
- Target: Prof. Mitra (already a connection via RSG), one Bio-X-affiliated faculty member
- Questions: "How do undergrads currently find your lab?" / "Would you want to be listed in a student-facing directory?" / "What do you wish students knew before emailing you?"

Talk to 1–2 non-RSG EE or CS freshmen:

- Purpose: Validate that problem extends beyond the founder's personal experience
- Questions: Core interview framework above

Phase 2: Manual MVP (This Weekend — 5 hours)

Build a Notion or Google Sheet database with:

Lab Information (20+ labs to start):

- Lab name + professor
- Department
- Contact email
- Research description (plain English, 2–3 sentences)
- Lab type (Wet / Dry / Computational / Hardware / Mixed)
- Currently accepting (Yes / No / Fall 2026 / Spring 2027)
- Experience required (None / Some CS or lab basics / Significant)
- Class year accepted (Freshmen OK / Sophomore+ / Junior+)
- International students eligible (Yes / No / Confirm per program)
- Paid or unpaid
- Typical commitment (Semester / Year / Multi-year)
- Hidden pipeline note ("Students from EE 108 often get recruited" / "Attend MSE dept. mixer")
- Last updated (date)

Distribution Plan:

- Share with Person 4 and Person 5 (both expressed strong intent)
- Post in Stanford EE, CS, Bio GroupMes and Slack channels
- Share in Splash, Frosh class Facebook group, dorm community boards
- Email to Bechtel International Center or international student community (for citizenship-eligibility feature)

Phase 3: Behavioral Testing (Week of Manual MVP — 1 week)

Critical Metrics:

- How many students view the database?
- How many email professors based on it? ← Most important
- Do students report back on responses?
- Do students share with friends unprompted?
- Does the Founder use it to find their next research opportunity?

Success Criteria to Proceed to Technical Build:

- 5+ students email professors based on database
- 2+ students get positive responses
- Students request additional labs be added
- At least 1 student shares with a friend unprompted

Phase 4: Technical Platform (Weeks 2–3)

If behavioral validation succeeds:

Tech Stack:

- Frontend: Next.js (React) + Tailwind CSS

- Database: Supabase (PostgreSQL) or Airtable
- Hosting: Vercel (free tier)
- Auth: None for MVP (public read access; editor access for updates)

MVP Feature Set (Validated):

- Keyword search across all fields
- Filter panel: department, lab type, accepting now, experience, international eligible, paid/unpaid, year accepted
- Lab cards: description, contact, all filter data, hidden pipeline notes, "last updated" date
- Mobile-responsive (students use phones)
- "Report outdated info" button

What NOT to Build in MVP:

- ❌ User accounts
- ❌ Application tracking
- ❌ Professor self-service portal (manual updates for MVP)
- ❌ Student reviews (V2)

Key Insights & Learnings

Insight 1: "Luck" dominates research access even at a research university

Four out of six people described successful research access as requiring luck, the right social connection, or stumbling into the right event. This is systemic, not individual failure.

Insight 2: Stanford has infrastructure, not UX

CURIS, Bio-X, MSE REU, SURP, Community Academic Profiles — the tools exist. But they're siloed, non-uniform, and require students to already know what they're looking for. The gap is discovery and aggregation, not the absence of programs.

Insight 3: Information asymmetry, not actual scarcity

Person 3 (RSG) confirmed that the lab is open to undergrads and says so on the website. Person 4 found that most bio professors do take students. The problem isn't lack of opportunity — it's that students have no way to see across the landscape.

Insight 4: International student eligibility is a hidden and painful barrier

CURIS PURE requires Federal Work Study (US citizens and qualifying non-citizens only). Many NSF-funded summer REUs are US-citizen-only. None of this is disclosed in a single accessible place. Students discover ineligibility after investing significant effort.

Insight 5: Day-to-day fit matters, not just discovery

Person 5 found research but might have chosen differently with better information about what the work actually involves. A platform that includes "what does working in this lab actually look like" would serve students better than one that only surfaces professor research abstracts.

Insight 6: Even the supply side (labs) would benefit

Person 3 noted that RSG's recruitment happens through EE 108 and email, and that many interested students don't reach out because they don't know the lab exists or how to approach it. A platform reduces inbound noise (generic emails) and increases quality (targeted outreach from students who understand the lab's work).

Insight 7: The hidden pipeline problem is structurally identical to Howard's

At Howard: General Bio Lab → summer research → Honors program. At Stanford: EE 108 → RSG; Bio lab course → Bio-X; MSE mixer → MSE lab. The mechanism is the same: access flows through specific classes and events that are never advertised as research entry points.

Document Prepared By: Lanon

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Next Milestone: Manual MVP database launch (target: this weekend)